

Social Media Performance Predictor – Multi-Brand Edition

Build a system that predicts Instagram post performance for beverage brands.

Context

You're given a dataset of ~375 Instagram posts across 5 beverage brands competing in the Indian market. The dataset includes: captions, media (video/images hosted on S3), engagement metrics (views, likes, comments, shares, engagement_rate), collaborators, posting metadata, and AI-generated visual summaries of the media.

The brands span different content strategies – sports sponsorships, pop culture, comedy, festivals, music events – and what "works" varies significantly across them.

Objective

Given any new post creative (image/video + caption + metadata), predict how it will perform – and explain why.

Your system should:

1. **Analyze** the dataset to understand what drives engagement
2. **Extract features** from post content (text, visual, metadata)
3. **Predict** whether a new post will perform well or poorly
4. **Explain** which factors are contributing to the prediction

How you architect this – whether you treat all brands as one model, build per-brand models, use namespaces, ensembles, or something else entirely – is up to you. Justify your choices.

Core Requirements

Prediction API

- Expose your system as an API
- Accept a new post draft (caption + image/video + any relevant metadata) as input
- Return a performance prediction with explanation
- Handle edge cases gracefully (missing data, unseen brand styles, expired media URLs, posts with no views because they're static images not reels, etc.)

Demo Frontend

- Build a simple frontend where a user can:
 - Upload or link a post creative (image + text, or video + text)
 - Select/specify the brand context
 - Get a prediction with explanation of what's driving it
- This doesn't need to be beautiful – it needs to work and clearly show the prediction + reasoning

Evaluation Strategy

This is critical. AI systems need rigorous evaluation to be trustworthy.

- **Design and document your evaluation approach.** How do you know your predictions are meaningful and not just noise?
- **Show concrete evaluation results.** Run your evaluation and present the output – metrics, examples, failure cases.
- **Consider at minimum:**
 - How you split data (given small N per brand)
 - What baseline you're comparing against (random? average? always-predict-medium?)
 - How you measure prediction quality
 - Where your system fails and why
 - How you'd improve evaluation with more data

We want to see your thinking on evaluation as much as the results themselves.

Edge Cases to Handle

- Some posts have views: 0 – this typically means it's a static post/album, not a reel. This is expected behavior, not missing data.
- Media URLs on S3 may be expired or inaccessible – your system should handle this gracefully
- Engagement rates vary wildly (0.05% to 100%+) due to follower count differences and content type dynamics
- Some posts may have collaborators with very different audience sizes – consider how this affects engagement attribution
- The dataset may contain noise or posts that don't belong to the brand's core content strategy – how you handle this is part of the evaluation
- A brand with very few posts should still produce a reasonable prediction (degrade gracefully, don't crash)

Evaluation Criteria

Criteria	Weight
Evaluation strategy & rigor	25%
Feature engineering creativity & depth	25%
Modeling approach & justification	20%
Explanation quality (interpretability)	15%
Code quality, API design & demo	15%

Deliverables

- A Python backend (framework of your choice) with the prediction API. Share a GitHub repo.
- A demo frontend (any framework) that lets a user upload a creative and get a prediction
- The dataset file will be provided: `assignment-dataset.json`
- A `README.md` describing:
 - How to run locally (backend + frontend)
 - Your architecture decisions and why
 - Your modeling approach and why you chose it
 - Your evaluation strategy, results, and analysis of failure cases
 - Key findings from your analysis of the dataset
 - What you'd do differently with 10x more data
- A 3-4 minute Loom video:
 - Demo the system working end-to-end (upload a creative → get prediction)
 - Walk through your evaluation approach and results
 - Explain your key architectural and modeling decisions

Hints & Non-Requirements

- You're free to augment the provided dataset with your own synthetic or real data — scrape more posts, generate synthetic examples, or add entirely new brands. The provided dataset is a starting point, not a constraint.
- You're not restricted to the brands in the dataset — configure your system to work with any brand, not just the five provided.
- You do NOT need to build a production ML model — a well-reasoned LLM-based approach, embedding similarity, or hybrid system is perfectly valid

- You may use any combination of: classical ML, LLM-based scoring, embedding similarity, rule-based scoring, few-shot prompting
- The `summary` field on media items already contains vision model output – use it
- Focus on reasoning and evaluation rigor more than raw prediction accuracy
- A simple approach done well beats a complex approach done poorly
- Think about what happens when you add a new brand with only 20 posts

Data Notes

- Dataset contains posts from: `cocacola_india`, `redbullindia`, `pepsiindia`, `sprite_india`, `thumsupofficial`
- Each post has an `_id` and a `data` object with all fields
- Media items include S3 URLs and a `summary` field with pre-computed vision model descriptions
- Not all posts have all fields – handling missing data is expected